

CYGNSS_L2_Python_Module_Demo

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1 CYGNSS L2 Python Module Demonstration

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1.0.1 Introduction

Welcome to the CYGNSS L2 Wind Python Module Demonstration. For the time being, I am calling this module `pygnss`. In addition to the `pygnss` source code and this notebook, I am also supplying the `cmap_map` module. This provides the ability to lighten or “wash out” a color table, which is useful when overplotting CYGNSS winds on top of the background winds. Put it in your `PYTHONPATH` along with `pygnss`.

OK, first things first, let’s import everything we need and prep our datasets.

```
In [1]: from __future__ import print_function
import warnings
warnings.filterwarnings('ignore')
import numpy as np
import matplotlib.pyplot as plt
import glob
import os
import pygnss
# cmap_map is also part of SingleDop (https://github.com/nasa/SingleDop)
# from singledop.cmap_map import lighten_cmap
from cmap_map import lighten_cmap
%matplotlib inline
```

The following cell will need to be updated by you. Input refers to the model or observed winds you ingested into the CYGNSS E2ES. Since the E2ES expects input data to be formatted a certain way, I exploited this to create Python objects to ingest and display these data separate from or alongside the synthetic CYGNSS data.

Output refers to the Level 2 output wind files created by the E2ES.

In the present example, my input data came from a limited 9-km resolution WRF domain, featuring a simulation of the atmosphere above the Indian Ocean during 11/25/2011. Off the coast of India was the developing Tropical Storm 05A. The CYGNSS E2ES was run on a day’s worth of WRF background simulation, with E2ES using a time step of 1 min. This is greatly decimated compared to CYGNSS’s native resolution of 1 second.

```
In [2]: input_dir = '/Users/tjlang/Documents/CYGNSS/E2ES/datafiles/winds/DYNAMO_WRF/'
input_files = sorted(glob.glob(input_dir+'CYGNSS_Sim_wrfout_d01_2011-11-25_*.nc'))
output_dir = '/Users/tjlang/Documents/CYGNSS/E2ES/run_test_mjo/output/'
output_L2_files = sorted(glob.glob(output_dir+'cyg?.L2.20111125-S000000-E235819.e2e.nc'))
print('E2ES Inputs', [os.path.basename(filen) for filen in input_files])
print()
print('E2ES Outputs', [os.path.basename(filen) for filen in output_L2_files])
```

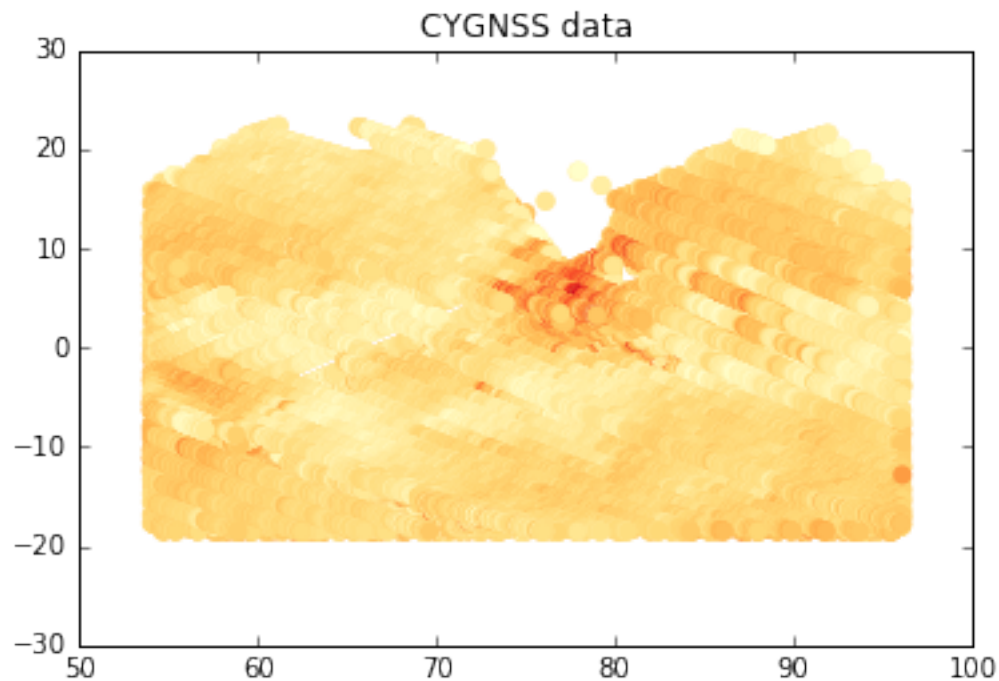
E2ES Inputs ['CYGNSS_Sim_wrfout_d01_2011-11-25_00:00:00.nc', 'CYGNSS_Sim_wrfout_d01_2011-11-25_01:00:00.nc']

E2ES Outputs ['cyg0.L2.20111125-S000000-E235819.e2e.nc', 'cyg1.L2.20111125-S000000-E235819.e2e.nc', 'cyg2.L2.20111125-S000000-E235819.e2e.nc']

1.0.2 Data Plotting

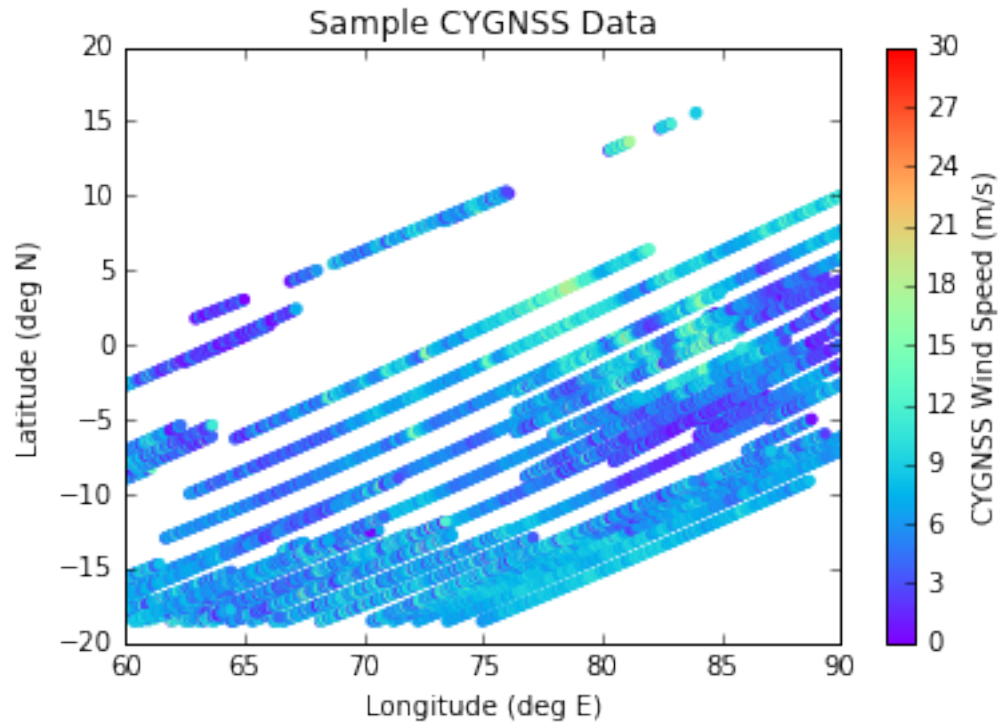
OK, ingesting and plotting the E2ES output data is as simple as 3 lines of code (actually, this can be reduced to two lines, as we'll see later).

```
In [3]: merged_data = pygnss.CygnssMultiSat(output_L2_files)
        display = pygnss.CygnssL2WindDisplay(merged_data)
        display.specular_plot()
```



Every specular point is a dot, with color corresponding to wind speed. Well, the above plot is a start. Why don't we spruce it up a bit though?

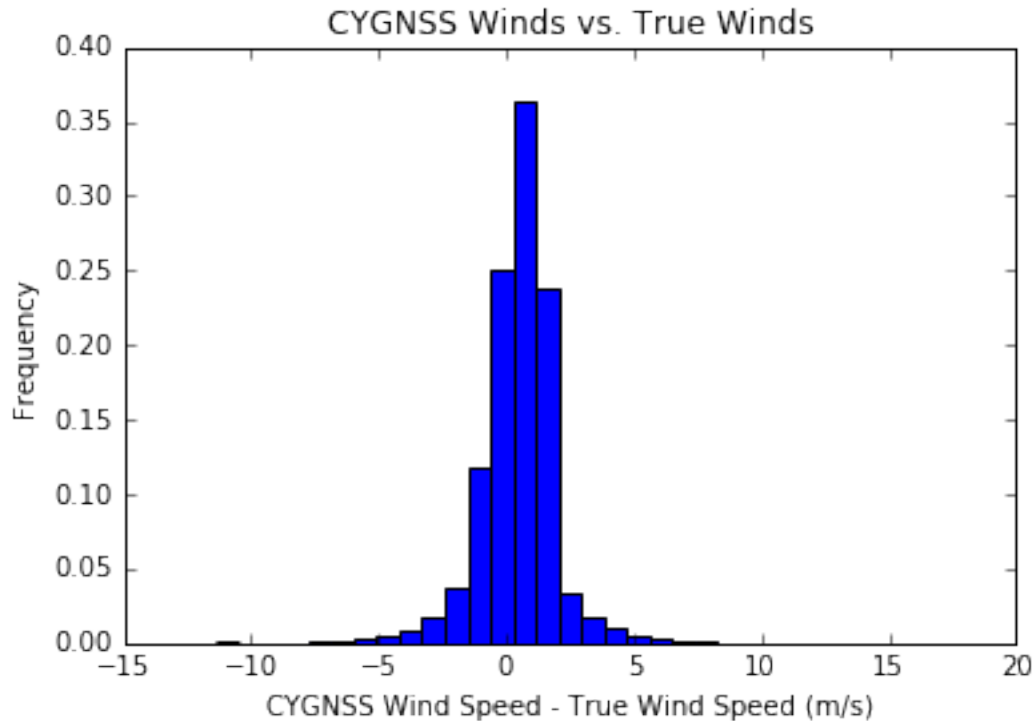
```
In [4]: display.specular_plot(colorbar_flag=True, axis_label_flag=True,
                               title='Sample CYGNSS Data', ms=20,
                               cmap='rainbow', indices=[10000, 15000],
                               latrange=[-20, 20], lonrange=[60, 90])
```



So the `specular_plot` method accepts keywords to adjust axes, title, colorbars/maps, marker size, time period (via the `indices` keyword), etc. See the method's docstring in the source code for more info.

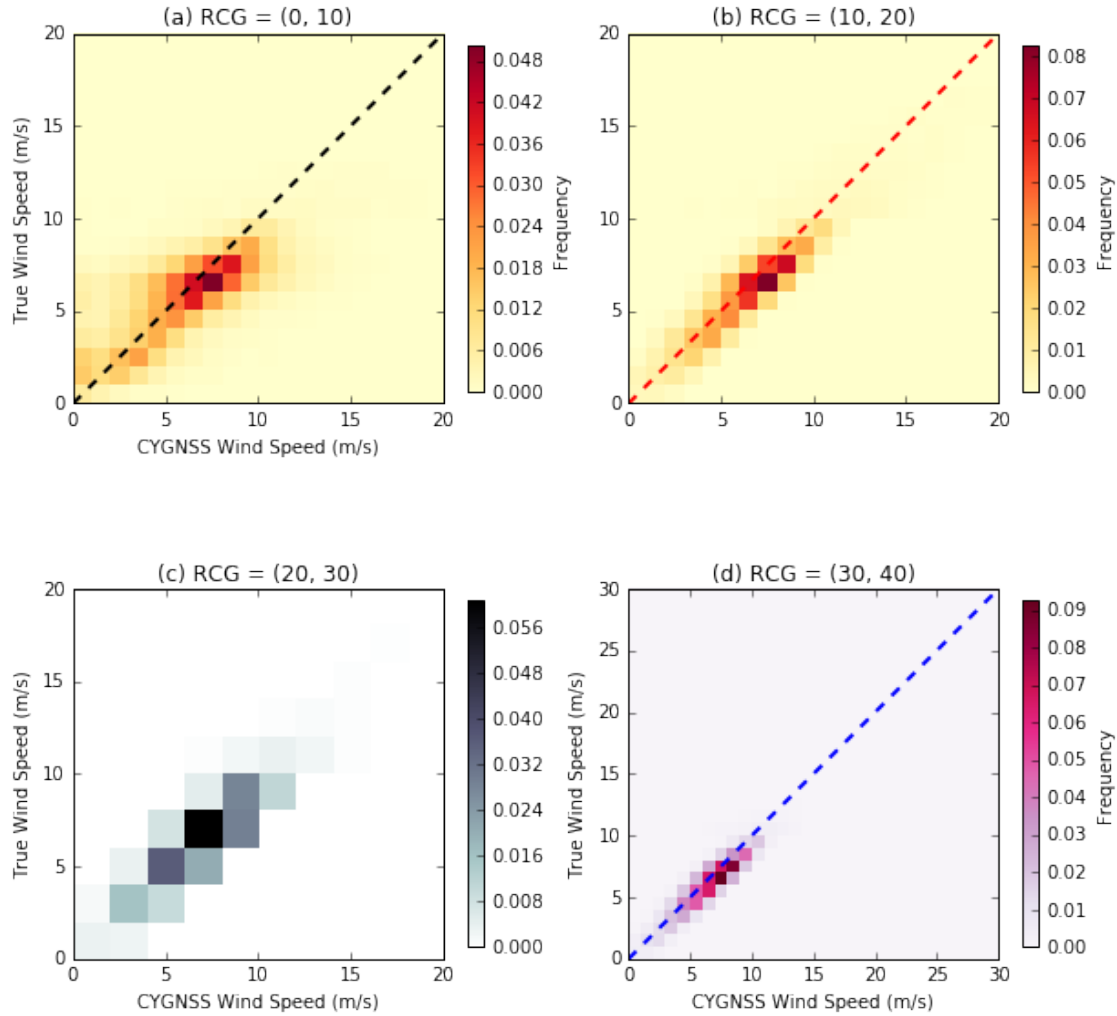
How about comparing synthetic CYGNSS versus “truth”? All it takes is one line of code to get a quick histogram.

```
In [5]: display.histogram_plot(bins=30, axis_label_flag=True)
```



You can also do a 2D histogram plot. Both histogram options including the ability to threshold on `RangeCorrectedGain` values via the `gain` keyword. There are lots of other keyword options too, for customizing the figure.

```
In [6]: fig, [[ax1, ax2], [ax3, ax4]] = plt.subplots(2, 2, figsize=(10, 10))
display.hist2d_plot(gain=[0, 10], cmap='YlOrRd', add_line=True,
                    line_color='k', ax=ax1, title='(a) RCG = (0, 10)',
                    axis_label_flag=True)
display.hist2d_plot(gain=[10, 20], ax=ax2, title='(b) RCG = (10, 20)')
display.hist2d_plot(gain=[20, 30], cmap='bone_r', add_line=False,
                    ax=ax3, title='(c) RCG = (20, 30)', axis_label_flag=True,
                    colorbar_label_flag=False, bins=10)
display.hist2d_plot(gain=[30, 40], bins=30, range=(0, 30), cmap='PuRd',
                    add_line=True, line_color='b', ax=ax4,
                    axis_label_flag=True, title='(d) RCG = (30, 40)')
```



We can also retrieve individual tracks of specular points, which are formed by reflections between one particular GPS satellite and a CYGNSS observatory. Simple box filtering of the wind speeds is also supported. The following example identifies all individual tracks and filters wind speeds along the tracks using a 5-pt filter.

```
In [7]: trl = pygnss.get_tracks(display, verbose=True, filter=True, window=5)
```

```
CYGNSS satellite 0
CYGNSS satellite 1
CYGNSS satellite 2
CYGNSS satellite 3
CYGNSS satellite 4
CYGNSS satellite 5
CYGNSS satellite 6
CYGNSS satellite 7
```

Now let's plot up a track and see how the raw and filtered wind speeds compare to truth. Filtering can greatly improve the error characteristics of CYGNSS data.

```
In [8]: import matplotlib.dates as mdates
        ind = 200
```

```

fig = plt.figure(figsize=(10, 5))
plt.plot(trl[ind].datetimes, trl[ind].ws, label='Raw', color='#F9D1CE')
plt.plot(trl[ind].datetimes[2:-2], trl[ind].fws, label='5-pt', color='#FF1A0A', lw=2)
plt.plot(trl[ind].datetimes, trl[ind].twc, label='True', color='#439FF0', lw=2)
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%H:%M:%S'))
plt.ylabel('Wind Speed (m/s)')
plt.xlabel('Time (UTC)')
plt.legend(loc='upper right')
plt.xticks(rotation=45)

```

```

//anaconda/lib/python2.7/site-packages/dateutil/rrule.py:470: DeprecationWarning: Using both 'count' and
"raise an error.", DeprecationWarning)

```

```

//anaconda/lib/python2.7/site-packages/dateutil/rrule.py:470: DeprecationWarning: Using both 'count' and
"raise an error.", DeprecationWarning)

```

```

Out[8]: (array([ 734466.16196759,  734466.16231481,  734466.16266204,
                  734466.16300926,  734466.16335648,  734466.1637037 ,
                  734466.16405093,  734466.16439815,  734466.16474537,
                  734466.16509259]), <a list of 10 Text xticklabel objects>)

```

```

//anaconda/lib/python2.7/site-packages/dateutil/rrule.py:470: DeprecationWarning: Using both 'count' and
"raise an error.", DeprecationWarning)

```

```

//anaconda/lib/python2.7/site-packages/dateutil/rrule.py:470: DeprecationWarning: Using both 'count' and
"raise an error.", DeprecationWarning)

```

```

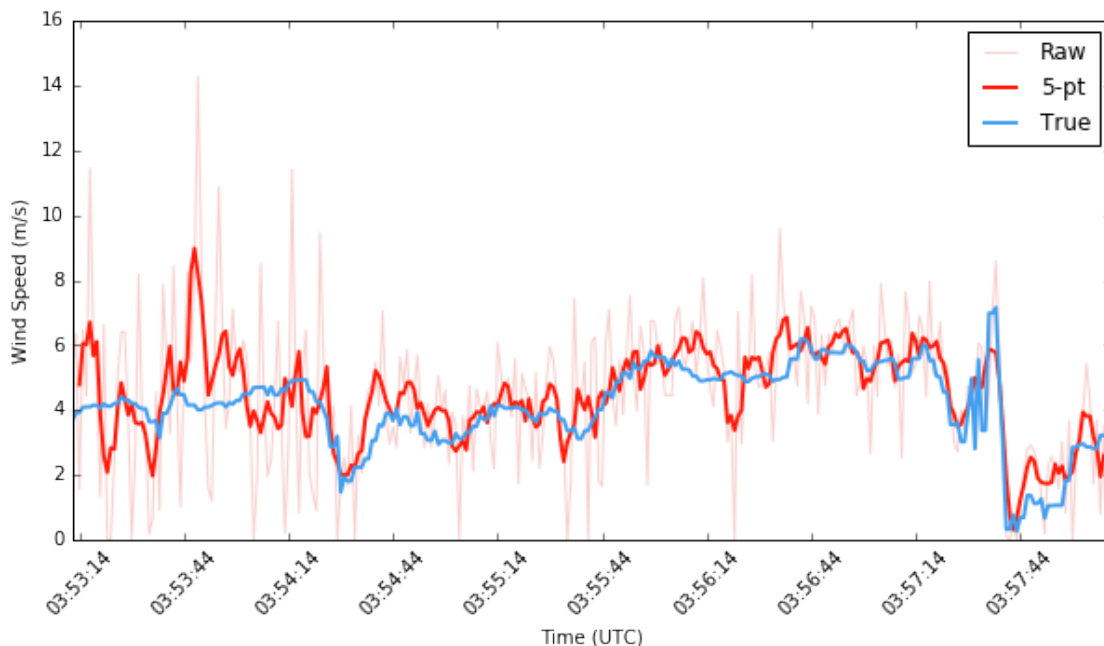
//anaconda/lib/python2.7/site-packages/dateutil/rrule.py:470: DeprecationWarning: Using both 'count' and
"raise an error.", DeprecationWarning)

```

```

//anaconda/lib/python2.7/site-packages/dateutil/rrule.py:470: DeprecationWarning: Using both 'count' and
"raise an error.", DeprecationWarning)

```



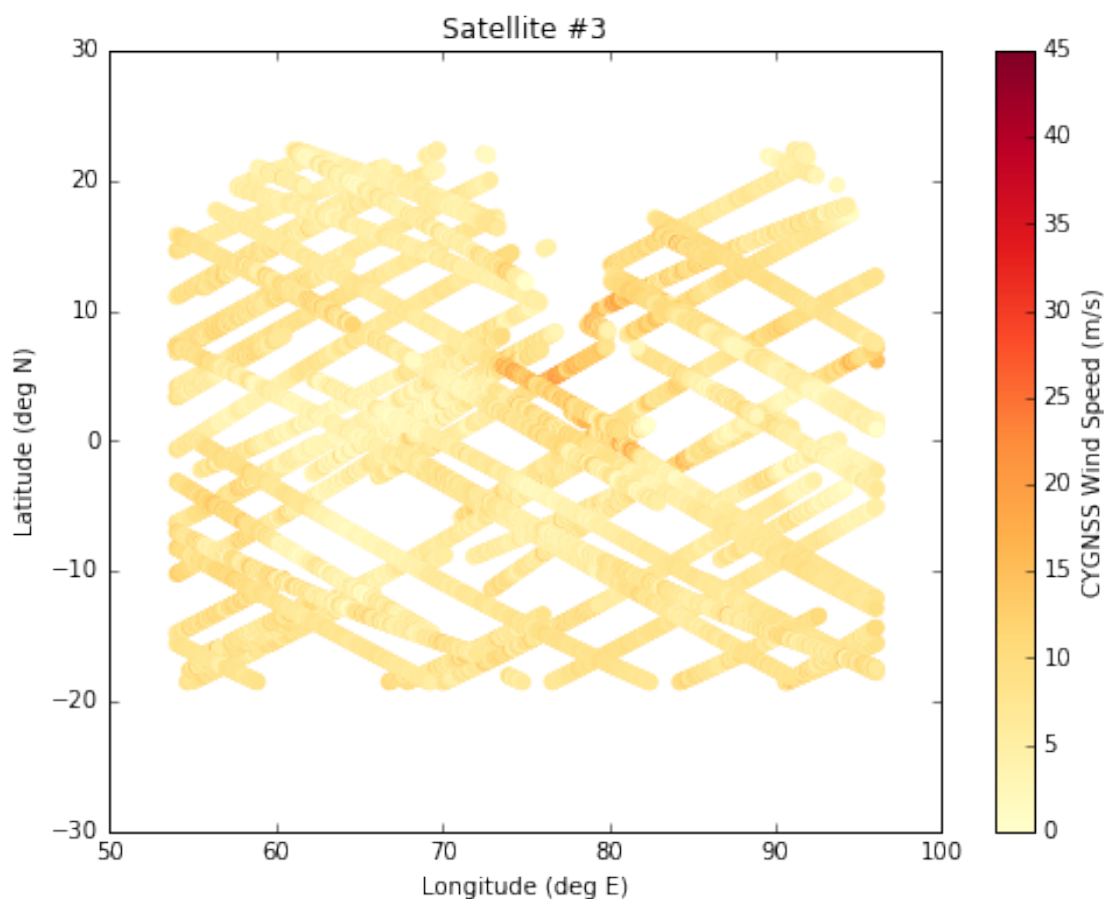
1.0.3 The structure of PyGNSS objects

So currently `pygnss` supports simple ways to get basic geospatial and statistical plots of the L2 wind data, to enable rapid analysis of E2ES output. But how does it actually work? Well, the design is object-oriented and modular, to make it easy to develop new functionality.

The core object is `NetcdfFile`, a simplified interface to ingest a netCDF file and obtain all of its variables in an organized manner. Then `CygnssSingleSat`, a child class of `NetcdfFile`, provides the actual interface to an individual satellite's data file. The `CygnssMultiSat` object makes use of multiple `CygnssSingleSat` objects to ingest the entire constellation's data and organize them appropriately. Finally, the `CygnssL2WindDisplay` object accepts either `CygnssSingleSat` or `CygnssMultiSat` objects (or a list of files from which such objects will be pulled) and gives access to the plotting routines seen above.

Let's get a better sense of how this works by investigating a single satellite's data.

```
In [9]: display = pygnss.CygnssL2WindDisplay(output_L2_files[2], number=8)
fig = plt.figure(figsize=(8, 6))
display.specular_plot(title='Satellite #3', colorbar_flag=True, axis_label_flag=True, vmax=45)
```



So, if all you want is a plot, we can directly ingest into a `CygnssL2WindDisplay` object an individual satellite's file, and the number keyword defines what number of specular points per time step to compare against the satellite object's `RangeCorrectedGain` attribute. The default is to take the specular points for the top-4 gain values (same as real-life CYGNSS). This is used to help mask out extra specular points. In addition, masking on bad or missing data is also employed when plotting the data.

Currently, land/ocean masking is not employed when plotting the data in this basic manner. That feature will be added soon. However, land can be easily masked when doing a `Basemap` plot, discussed later.

OK, enough plotting, what is actually in these objects?

```
In [10]: sat3 = pygnss.CygnssSingleSat(output_L2_files[2])
         print(sat3.variable_list)

[u'Year', u'Month', u'Day', u'Hour', u'Minute', u'Second', u'GpsID', u'Latitude', u'Longitude', u'WindSpeed']

In [11]: print(np.shape(sat3.WindSpeed))

(86300, 15)
```

So all the stuff in the L2 wind files is there, and the shape for many key variables like `WindSpeed` is (number of time steps, number of possible specular points = 15). Masking is simply used later to cut that down to 4 specular points per time step, or whatever the end user wants.

```
In [12]: sat3.get_gain_mask(number=8)
         print(sat3.variable_list)

[u'Year', u'Month', u'Day', u'Hour', u'Minute', u'Second', u'GpsID', u'Latitude', u'Longitude', u'WindSpeed']

In [13]: index = 1000
         print(sat3.GoodData[index], sat3.RangeCorrectedGain[index])

[0 0 0 0 0 0 0 1 1 1 1 1 1 1 1] [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
```

In this case, the top-8 specular points in terms of `RangeCorrectedGain` at time step = index get `GoodData` flags = 1. If you just ingest a file using `CygnssSingleSat`, the gain mask (the `GoodData` attribute) is not applied automatically. However, the `CygnssL2WindDisplay` and `CygnssMultiSat` objects will set this automatically based on user input (or lack thereof), so that the plotting routines will know what to plot.

Indeed, the `CygnssMultiSat` object has everything in `CygnssSingleSat` too, but consolidated to fit data from all the satellites into one object. It even holds all the individual `CygnssSingleSat` objects used to construct it within its `satellites` attribute.

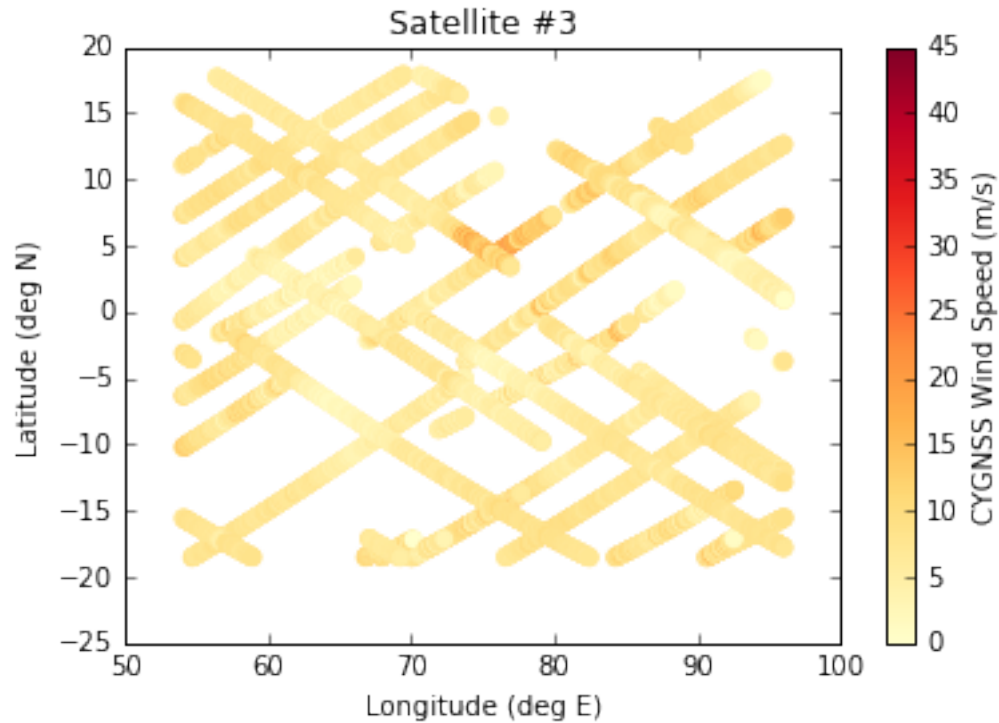
```
In [14]: print(merged_data.variable_list)
         print(np.shape(merged_data.WindSpeed))

[u'Year', u'Month', u'Day', u'Hour', u'Minute', u'Second', u'GpsID', u'Latitude', u'Longitude', u'WindSpeed']
(86300, 120)

In [15]: print(merged_data.satellites)

[<pygnss.CygnssSingleSat object at 0x106384610>, <pygnss.CygnssSingleSat object at 0x1063847d0>, <pygnss.CygnssSingleSat object at 0x106384920>]

In [16]: display = pygnss.CygnssL2WindDisplay(merged_data.satellites[2])
         display.specular_plot(title='Satellite #3', colorbar_flag=True,
                               axis_label_flag=True, vmax=45)
```

And now we basically recreated the figure from above, working from the `merged_data` object itself, although with only the top-4 specular points in terms of `RangeCorrectedGain` plotted (the default when we created `merged_data` several cells back).

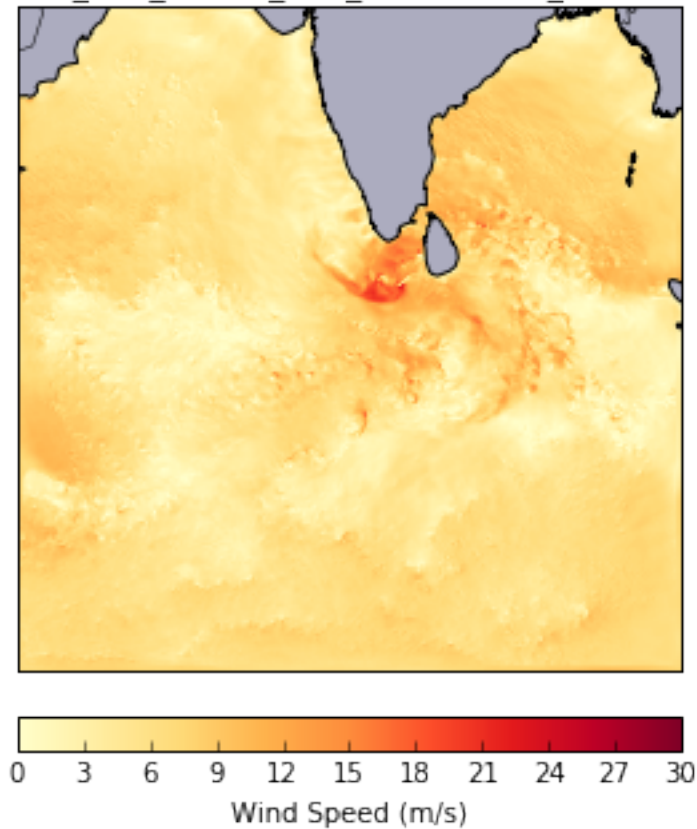
1.0.4 Plotting E2ES input data along with CYGNSS on a Basemap

The `pygnss` module also can work with the E2ES input data files. The key objects here are `E2esInputData` (for interfacing with the data file) and `InputWindDisplay` (for helping plot the data).

```
In [17]: input_data = pygnss.E2esInputData(input_files[11])
         fig = plt.figure(figsize=(5,5))
         input_display = pygnss.InputWindDisplay(input_data)
         input_display.basemap_plot(title=os.path.basename(input_files[11]))
```

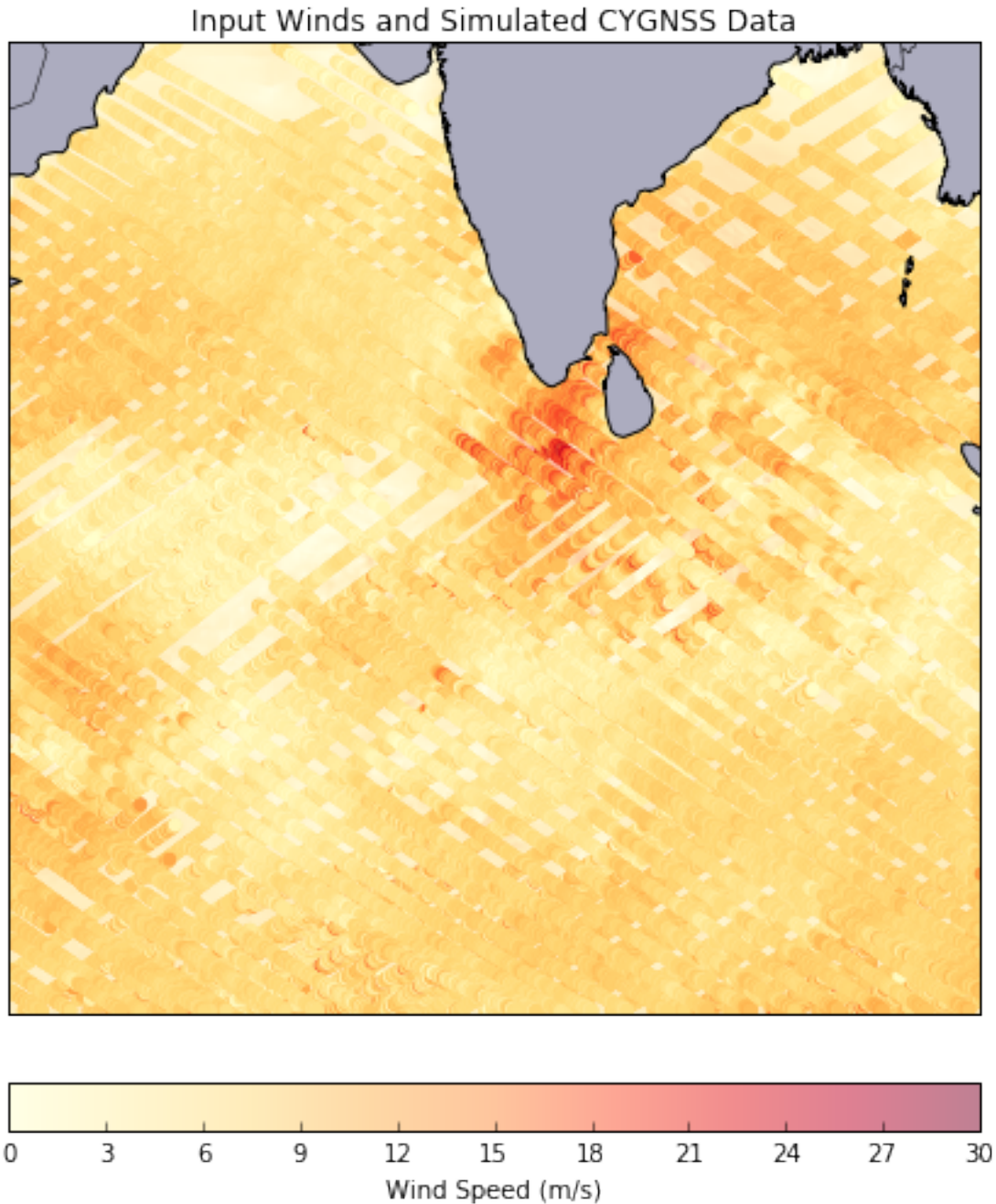
```
Out[17]: <matplotlib.basemap.Basemap at 0x14d5e8490>
```

CYGNSS_Sim_wrfout_d01_2011-11-25_11:00:00.nc



Note the output Basemap object. This is by default returned automatically by the `basemap_plot` method. We can take advantage of this to bring in the CYGNSS data, like so:

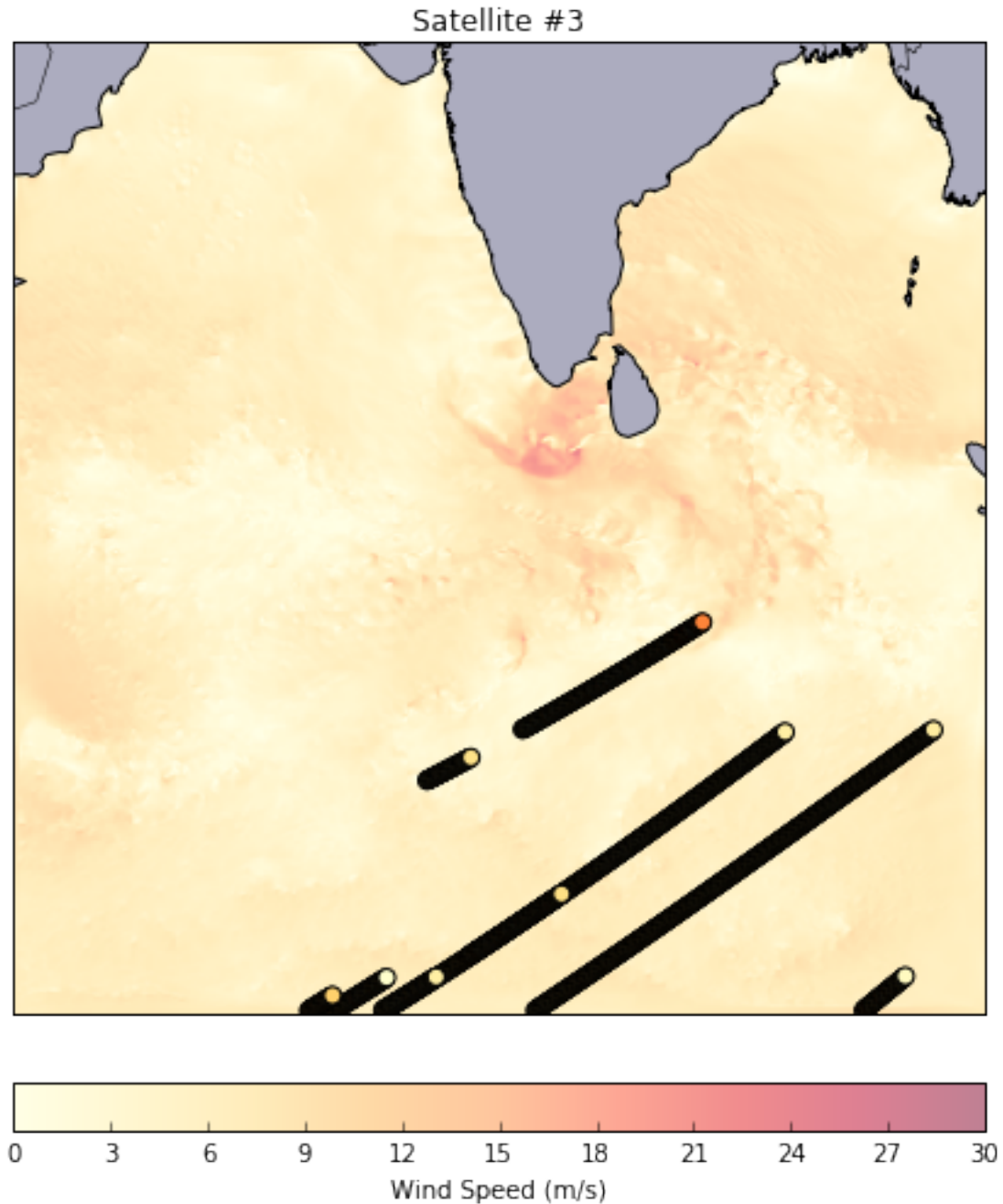
```
In [18]: lgist = lighten_cmap(plt.cm.get_cmap('YlOrRd'))
fig = plt.figure(figsize=(8, 8))
input_display = pygnss.InputWindDisplay(input_files[11])
m = input_display.basemap_plot(title='Input Winds and Simulated CYGNSS Data', cmap=lgist)
display = pygnss.CygnssL2WindDisplay(output_L2_files)
display.specular_plot(colorbar_flag=False, axis_label_flag=False, title_flag=False,
                      basemap=m, edge_flag=False, cmap='YlOrRd', ms=35)
```



Note how `InputWindDisplay` can accept files directly, bypassing the `E2esInputData` step. I also lightened the base colormap so the CYGNSS specular points show up a bit better.

And you can plot the data for a single satellite, limit by time, color in the edges of the specular points, etc.

```
In [19]: fig = plt.figure(figsize=(8,8))
          m = input_display.basemap_plot(title='Satellite #3', cmap=lgist)
          ss_display = pygnss.CygnssL2WindDisplay(merged_data.satellites[2])
          ss_display.specular_plot(cmap='YlOrRd', colorbar_flag=False, edge_flag=True,
                                  title_flag=False, basemap=m, indices=[5000, 10000])
```



You can use a different `Basemap` than the one created by `InputWindDisplay.basemap_plot` to plot CYGNSS data, if you wish. The `specular_plot` syntax above would be the same. Below is an example of how to modify the `Basemap` after the fact.

```
In [20]: lgist = lighten_cmap(plt.cm.get_cmap('YlOrRd'))
fig = plt.figure(figsize=(8, 8))
input_display = pygnss.InputWindDisplay(input_files[11])
m = input_display.basemap_plot(title='Input Winds and Simulated CYGNSS Data', cmap=lgist,
                                return_flag=True)
```

```

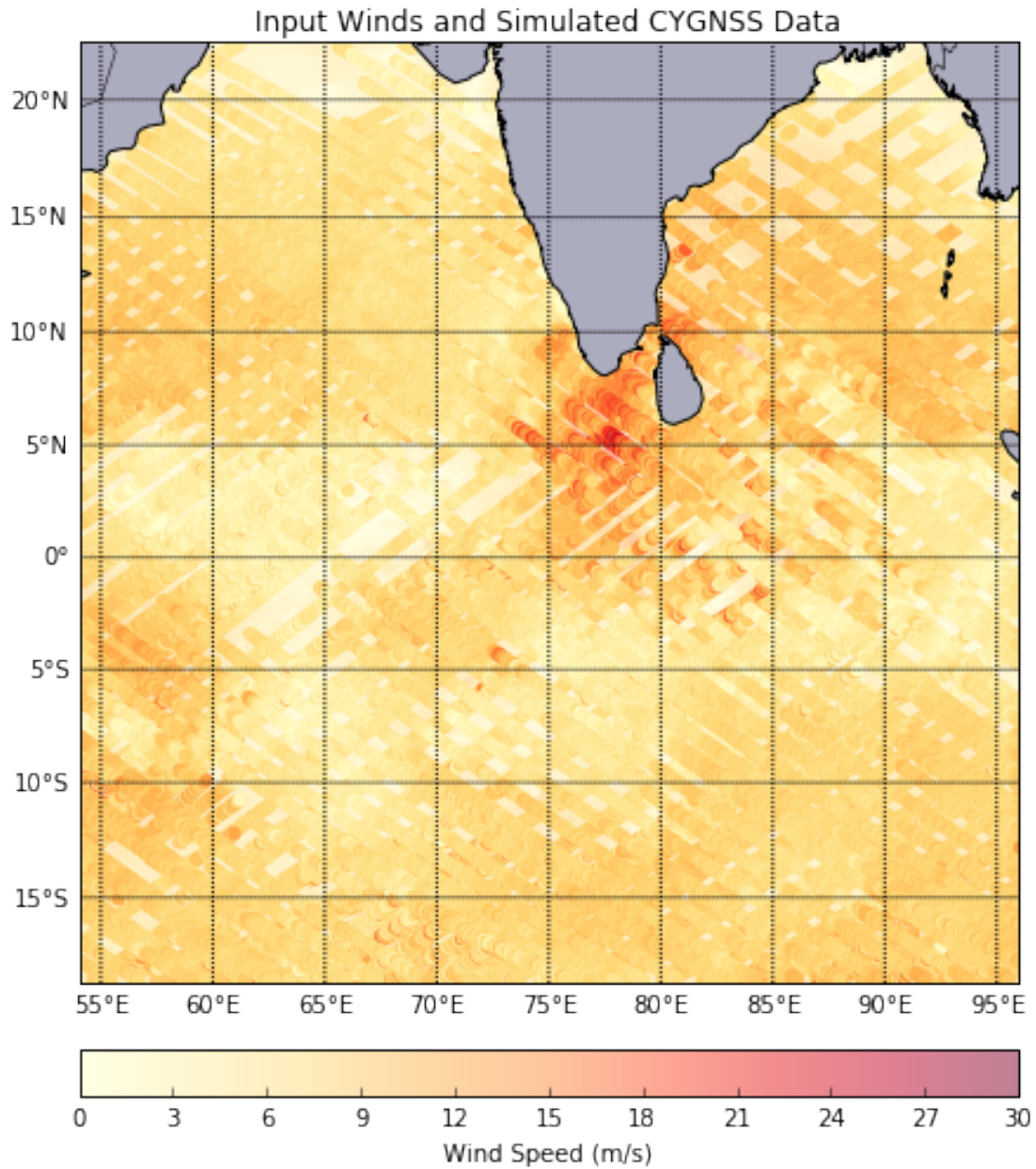
display = pygnss.CygnssL2WindDisplay(output_L2_files)
display.specular_plot(
    colorbar_flag=False, axis_label_flag=False, title_flag=False,
    basemap=m, edge_flag=False, cmap='YlOrRd', ms=35)
m.drawmeridians(np.arange(-180, 180, 5), labels=[0, 0, 0, 1])
m.drawparallels(np.arange(-90, 90, 5), labels=[1, 0, 0, 0])

```

```

Out[20]: {-15: ([<matplotlib.lines.Line2D at 0x1414f1f50>],
  [<matplotlib.text.Text at 0x47518f150>]),
-10: ([<matplotlib.lines.Line2D at 0x47517ac90>],
  [<matplotlib.text.Text at 0x47518f090>]),
-5: ([<matplotlib.lines.Line2D at 0x47517a990>],
  [<matplotlib.text.Text at 0x475184d90>]),
0: ([<matplotlib.lines.Line2D at 0x4751a4150>],
  [<matplotlib.text.Text at 0x4751844d0>]),
5: ([<matplotlib.lines.Line2D at 0x4751a4690>],
  [<matplotlib.text.Text at 0x475184a50>]),
10: ([<matplotlib.lines.Line2D at 0x17c04e850>],
  [<matplotlib.text.Text at 0x473b63190>]),
15: ([<matplotlib.lines.Line2D at 0x17c04ead0>],
  [<matplotlib.text.Text at 0x473b63550>]),
20: ([<matplotlib.lines.Line2D at 0x47518ff90>],
  [<matplotlib.text.Text at 0x473b637d0>])}

```

And that's about it. The rest of this notebook is sparsely documented, but provides examples of how to make other figures, investigate the input and L2 data, etc.

```
In [21]: # Remember, you can always ...
         help(pygnss)
```

Help on module pygnss:

```
NAME
    pygnss
```

```
FILE
```

/Users/tjlang/Documents/Python/PYGNSS/pygnss.py

DESCRIPTION

Title/Version

Python CYGNSS Toolkit (PyGNSS)

pygnss v0.7

Developed & tested with Python 2.7 and 3.4

Author

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Overview

This module enables the ingest, analysis, and plotting of Cyclone Global Navigation Satellite System (CYGNSS) End-to-End Simulator (E2ES) input and output data. To use, place in PYTHONPATH and use the following import command:
import pygnss

Notes

Requires - numpy, matplotlib, Basemap, netCDF4, warnings, os, six, datetime, sklearn, copy

Change Log

v0.7 Major Changes (04/20/2016)

1. Added CygnssSubsection class to consolidate and simplify data subsectioning. Can be called independently, but is also used by the plotting methods in CygnssL2WindDisplay class. Moved the subsection_data and get_good_data_mask methods to this new class, and greatly modified them to eliminate method returns.
2. Added ability to subsection data by CYGNSS and GPS satellite numbers, as well as by range-corrected gain threshold or interval. All plotting routines now support these subsectioning capabilities.
3. Added get_datetime independent function to derive datetime objects in the same array shape as the WindSpeed data. Added datetime module dependency.
4. Added CygnssTrack class to leverage CygnssSubsection to help isolate and contain all the data from an individual track. Also enables filtering of wind speeds along a track.
5. Added get_tracks independent function to isolate all individual tracks, and return a list of CygnssTrack objects from a CygnssSingleSat, CygnssMultiSat, or CygnssL2WindDisplay object.

v0.6 Major Changes (11/20/2015)

1. Added hist2d_plot method to CygnssL2WindDisplay.

2. Added threshold keyword to allow filtering of histogram figures by RangeCorrectedGain windows.

v0.5 Major Changes (08/10/2015)

1. Supports Python 3 now.

v0.4 Major Changes (07/02/2015)

1. Made all code pep8 compliant.

v0.3 Major Changes (03/19/2015)

1. Documentation improvements. Doing help(pygnss) should be more useful now.
2. Fixed bug where GoodData attribute was not getting set for CygnssSingleSat objects after they were input into CygnssL2WindDisplay.
3. Fixes to ensure CygnssSingle/MultiSat classes can ingest L1 DDM files w/out errors. This provides a basis for adding L1 DDM functionality to PyGNSS.
4. Swapped out np.rank for np.ndim due to annoying deprecation warnings.

v0.2 Major Changes (02/24/2015)

1. Fixed miscellaneous bugs related to data subsectioning and plotting.
2. Added histogram plot for CYGNSS vs. Truth winds.

v0.1 Functionality:

1. Reads netCDFs for input to CYGNSS E2ES as well as output files.
2. Can ingest single-satellite data or merge all 8 together.
3. Capable of masking L2 wind data by RangeCorrectedGain.
4. Basic display objects & plotting routines exist for input/output data, w/ support for combined input/output plots.

Planned Updates

1. Enable subsectioning of output data specifically by time rather than index.
2. Support for DDM file analysis/plotting
3. Merged input/output display object for 1-command combo plots given proper inputs.
4. Get CygnssL2WindDisplay.specular_plot() to automatically adjust size of points to reflect actual CYGNSS spatial resolution on Basemap. Right now, user just has manual control of the marker size and would need to guess at this if they want the specular points to be truly spatially accurate.
5. Incorporate land/ocean flag in non-Basemap CYGNSS plots

CLASSES

```
__builtin__.object
    CygnssL2WindDisplay
    CygnssMultiSat
    CygnssSubsection
    CygnssTrack
    InputWindDisplay
    NetcdfFile
    CygnssSingleSat
    E2esInputData
```

```
class CygnssL2WindDisplay(__builtin__.object)
    | This display class provides an avenue for making plots from CYGNSS L2 wind
```



```
data.
```

```
Methods defined here:
```

```
__init__(self, cygnss_sat_object, number=4)
```

```
    cygnss_sat_object = CygnssSingle/MultiSat object, single L2 file,  
                        or list of files.
```

```
    number = Number of specular points to consider in RangeCorrectedGain  
             rankings.
```

```
hist2d_plot(self, title='CYGNSS Winds vs. True Winds', fig=None, ax=None, axis_label_flag=False,
```

```
    Plots a normalized 2D histogram of CYGNSS wind speed vs. true wind spd  
    (as provided by the input data to the E2ES). This information can be  
    thresholded by RangeCorrectedGain
```

```
    bins = Number of bins to use in the histogram
```

```
    title = Title of plot
```

```
    bad = Bad value of Lat/Lon to throw out
```

```
    fig = matplotlib Figure object to use
```

```
    ax = matplotlib Axes object to use
```

```
    axis_label_flag = Set to True to label lat/lon axes
```

```
    title_flag = Set to False to suppress title
```

```
    indices = Indices (2-element tuple) to use to limit the period of data  
             shown (i.e., limit by time)
```

```
    gain = Threshold for range-corrected gain (RCG). Can be 2-ele. tuple.
```

```
        If so, use only RCG within that range. If scalar, then use  
        data above the given RCG value.
```

```
    save = Name of image file to save plot to
```

```
    sat = CYGNSS satellite number (0-7)
```

```
    **kwargs = Whatever else pyplot.hist2d will accept
```

```
histogram_plot(self, title='CYGNSS Winds vs. True Winds', fig=None, ax=None, axis_label_flag=False,
```

```
    Plots a normalized histogram of CYGNSS wind speed vs. true wind speed  
    (as provided by the input data to the E2ES).
```

```
    bins = Number of bins to use in the histogram
```

```
    title = Title of plot
```

```
    bad = Bad value of Lat/Lon to throw out
```

```
    fig = matplotlib Figure object to use
```

```
    ax = matplotlib Axes object to use
```

```
    axis_label_flag = Set to True to label lat/lon axes
```

```
    title_flag = Set to False to suppress title
```

```
    indices = Indices (2-element tuple) to use to limit the period of data  
             shown (i.e., limit by time)
```

```
    gain = Threshold for range-corrected gain (RCG). Can be 2-ele. tuple.
```

```
        If so, use only RCG within that range. If scalar, then use  
        data above the given RCG value.
```

```
    save = Name of image file to save plot to
```

```
    sat = CYGNSS satellite number (0-7)
```

```
specular_plot(self, cmap='YlOrRd', title='CYGNSS data', vmin=0, vmax=30, ms=50, marker='o', bad=
```

```
    Plots CYGNSS specular points on lat/lon axes using matplotlib's scatter  
    object, which colors each point based on its wind speed value.
```

```

|     cmap = matplotlib or user-defined colormap
|     title = Title of plot
|     vmin = Lowest wind speed value to display on color table
|     vmax = Highest wind speed value to display on color table
|     ms = Size of marker used to plot each specular point
|     marker = Marker shape to use ('o' is best)
|     bad = Bad value of Lat/Lon to throw out
|     fig = matplotlib Figure object to use
|     ax = matplotlib Axes object to use
|     colorbar_flag = Set to True to show the colorbar
|     basemap = Basemap object to use in plotting the specular points
|     edge_flag = Set to True to show a black edge to make each specular
|                 point more distinctive
|     axis_label_flag = Set to True to label lat/lon axes
|     title_flag = Set to False to suppress title
|     indices = Indices (2-element tuple) to use to limit the period of data
|                 shown (i.e., limit by time)
|     save = Name of image file to save plot to
|     lonrange = 2-element tuple to limit longitude range of plot
|     latrange = 2-element tuple to limit latitude range of plot
|     gpsid = Integer ID number for GPS satellite to examine
|     sat = CYGNSS satellite number (0-7)
|     gain = Threshold for range-corrected gain (RCG). Can be 2-ele. tuple.
|             If so, use only RCG within that range. If scalar, then use
|             data above the given RCG value.
|     return_flag = Set to True to return Figure, Axes, and Basemap objects
|                     (in that order)
|
| -----
| Data descriptors defined here:
|
| __dict__
|     dictionary for instance variables (if defined)
|
| __weakref__
|     list of weak references to the object (if defined)
|
class CygnssMultiSat(__builtin__.object)
| Can ingest both L2 Wind and L1 DDM files. Merges the CYGNSS constellation's
| individual satellites' data together into a class structure very similar to
| CygnssSingleSat, just with bigger array dimensions.
|
| Methods defined here:
|
| __init__(self, l2list, number=4)
|     l2list = list of CygnssL2SingleSat objects or files
|     number = Number of maximum RangeCorrectedGain slots to consider
|
| merge_cygnss_data(self)
|     Loop over each satellite and append its data to the master arrays
|
| -----
| Data descriptors defined here:
|

```

```

|  __dict__
|      dictionary for instance variables (if defined)
|
|  __weakref__
|      list of weak references to the object (if defined)
|
class CygnssSingleSat(NetcdfFile)
|  Child class of NetcdfFile. Can ingest both L2 Wind and DDM files.
|  All variables within the files are incorporated as attributes of the
|  class. This class forms the main building block of PyGNSS.
|
|  Method resolution order:
|      CygnssSingleSat
|      NetcdfFile
|      __builtin__.object
|
|  Methods defined here:
|
|  get_gain_mask(self, number=4)
|      With L2 wind data, identifies top specular points in terms of
|      range corrected gain. Creates the GoodData attribute, which provides
|      a mask that analysis and plotting routines can use to only consider
|      specular points with the highest RangeCorrectedGain
|      number = Number of specular points to consider in the rankings
|
|  -----
|  Methods inherited from NetcdfFile:
|
|  __init__(self, filename=None)
|
|  fill_variables(self, volume)
|      Loop thru all variables and store them as attributes
|
|  read_netcdf(self, filename)
|      variable_list = holds all the variable key strings
|
|  -----
|  Data descriptors inherited from NetcdfFile:
|
|  __dict__
|      dictionary for instance variables (if defined)
|
|  __weakref__
|      list of weak references to the object (if defined)
|
class CygnssSubsection(__builtin__.object)
|  Class to handle subsectioning CYGNSS data. Subsectioning by
|  satellite (via CygnssSingleSat input), time indices, GPS satellite ID,
|  range-corrected gain, etc. is supported.
|
|  Main Attributes
|  -----
|  ws = Wind speed array
|  lon = Longitude array

```

```

| lat = Latitude array
| gd = GoodData array
| rcg = RangeCorrectedGain array
| gps = GpsID array
|
| Methods defined here:
|
| __init__(self, data, indices=None, gpsid=None, gain=None, bad=-500, sat=None)
|     data = CygnssSingleSat, CygnssMultiSat, or CygnssL2WindDisplay object
|     gpsid = Integer ID number for GPS satellite to examine
|     gain = Threshold by range-corrected gain, values below will be masked
|     bad = Value to compare against lat/lon to mask out missing data
|     sat = CYGNSS satellite number (0-7)
|     indices = Indices (2-element tuple) to use to limit the period of data
|               shown (i.e., limit by time)
|
| get_good_data_mask(self)
|     Sets a mask used to limit the data plotted. Filtered out are data
|     masked out by the GoodData mask (based on RangeCorrectedGain), missing
|     lat/lon values, and bad data (ws < 0)
|
| subsection_data(self)
|     This method subsections the L2 wind data and returns these as arrays
|     ready to plot.
|
| -----
| Data descriptors defined here:
|
| __dict__
|     dictionary for instance variables (if defined)
|
| __weakref__
|     list of weak references to the object (if defined)
|
| -----
| Data and other attributes defined here:
|
| __slotnames__ = []
|
class CygnssTrack(__builtin__.object)
| Class to facilitate extraction of a single track of specular points
| from a CygnssSingleSat, CygnssMultiSat, or CygnssL2WindDisplay object.
|
|
| Attributes
| -----
| input = CygnssSubsection object
| ws = CYGNSS wind speeds
| tws = Truth wind speeds
| lon = Longitudes of specular points
| lat = Latitudes of specular points
| rcg = Range-corrected gains of specular points
| datetimes = Datetime objects for specular points
|

```

```

| The following attributes are created by filter_track method:
| fws = Filtered wind speeds
| flon = Filtered longitudes
| flat = Filtered latitudes
| These attributes are shorter than the main attributes by the window length
|
| Methods defined here:
|
| __init__(self, data, datetimes=None, **kwargs)
|     data = CygnssSingleSat, CygnssMultiSat, or CygnssL2WindDisplay object
|     datetimes = List of datetime objects from get_datetime function.
|                 If None, this function is called.
|
| filter_track(self, window=5)
|     Applies a running-mean filter to the track.
|
|     window = Number of specular points in the running mean window.
|               Must be odd.
|
| -----
| Data descriptors defined here:
|
| __dict__
|     dictionary for instance variables (if defined)
|
| __weakref__
|     list of weak references to the object (if defined)
|
| -----
| Data and other attributes defined here:
|
| __slotnames__ = []
|
class E2esInputData(NetcdfFile)
| Base class for ingesting E2ES input data. Child class of NetcdfFile.
|
| Method resolution order:
|     E2esInputData
|     NetcdfFile
|     __builtin__.object
|
| Methods defined here:
|
| get_wind_speed(self)
|     Input E2ES data normally don't have wind speed as a field. This method
|     fixes that.
|
| -----
| Methods inherited from NetcdfFile:
|
| __init__(self, filename=None)
|
| fill_variables(self, volume)
|     Loop thru all variables and store them as attributes

```

```

| read_netcdf(self, filename)
|     variable_list = holds all the variable key strings
|
| -----
| Data descriptors inherited from NetcdfFile:
|
| __dict__
|     dictionary for instance variables (if defined)
|
| __weakref__
|     list of weak references to the object (if defined)
|
class InputWindDisplay(__builtin__.object)
| Display object for the E2ES input data
|
| Methods defined here:
|
| __init__(self, input_winds_object)
|     input_winds_object = Input E2esInputData object or wind file
|
| basemap_plot(self, fill_color='#ACACBF', ax=None, fig=None, time_index=0, cmap='YlOrRd', vmin=0,
|     Plots E2ES input wind speed data on a Basemap using matplotlib's
|     pcolormesh object. Defaults to return the Basemap object so other
|     things (e.g., CYGNSS data) can be overplotted.
|
|     fill_color = Color to fill continents
|     time_index = If the input data contain more than one time step, this
|                   index selects the time step to display
|     cmap = matplotlib or user-defined colormap
|     title = Title of plot
|     vmin = Lowest wind speed value to display on color table
|     vmax = Highest wind speed value to display on color table
|     fig = matplotlib Figure object to use
|     ax = matplotlib Axes object to use
|     return_flag = Set to False to suppress Basemap object return
|     title_flag = Set to False to suppress title
|     save = Name of image file to save plot to
|     colorbar_flag = Set to False to suppress the colorbar
|     show_grid = Set to True to show the lat/lon grid and label it
|
| make_coordinates_2d(self)
|
| -----
| Data descriptors defined here:
|
| __dict__
|     dictionary for instance variables (if defined)
|
| __weakref__
|     list of weak references to the object (if defined)
|
class NetcdfFile(__builtin__.object)
| Base class used for reading netCDF-format L1 and L2 CYGNSS data files

```

```

|
| Methods defined here:
|
| __init__(self, filename=None)
|
| fill_variables(self, volume)
|     Loop thru all variables and store them as attributes
|
| read_netcdf(self, filename)
|     variable_list = holds all the variable key strings
|
| -----
| Data descriptors defined here:
|
| __dict__
|     dictionary for instance variables (if defined)
|
| __weakref__
|     list of weak references to the object (if defined)

```

FUNCTIONS

```

check_for_strings(var)
    Given an input var, check to see if it is a string (scalar or array of
    strings), or something else.

    Output:
    0 = non-string, 1 = string scalar, 2 = string array

get_basemap(latrange=[-90, 90], lonrange=[-180, 180], resolution='1', area_thresh=1000)
    Function to create a specifically formatted Basemap provided the input
    parameters.

    latrange = Latitude range of the plot (2-element tuple)
    lonrange = Longitude range of the plot (2-element tuple)
    resolution = Resolution of the Basemap
    area_thresh = Threshold (in km**2) for displaying small features, such as
                  lakes/islands

get_datetime(data)

get_tracks(data, indices=None, min_samples=10, verbose=False, filter=False, window=5)
    Returns a list of CygnssTrack objects from a CYGNSS data or display object

    data = CygnssSingleSat, CygnssMultiSat, or CygnssL2WindDisplay object
    indices = Indices (2-element tuple) to use to limit the period of data
              shown (i.e., limit by time). Not usually necessary unless
              processing more than one day's worth of data.
    min_samples = Minimum allowable track size (number of specular points)
    verbose = Set to True for some text updates while running
    filter = Set to True to filter each track
    window = Window length of filter, in number of specular points. Must be odd.

parse_fig_ax(fig, ax)
    Parse matplotlib Figure and Axes objects, if provided, or just grab the

```


current ones in memory.

warn(...)

Issue a warning, or maybe ignore it or raise an exception.

DATA

VERSION = '0.7'

print_function = _Feature((2, 6, 0, 'alpha', 2), (3, 0, 0, 'alpha', 0)...

string_types = (<type 'basestring'>,))

1.0.5 Other plotting examples

Below is how I made the individual frames for the animated GIF I showed on the CYGNSS Science Team telecon.

```
In [22]: def plot_animation(display, multiple=60):
maxstep = 1 + np.shape(merged_data.WindSpeed)[0] / multiple
for i in np.arange(maxstep):
    plt.close()
    index_start = multiple*i
    index_stop = multiple*(i+1)
    print(index_start, index_stop)
    val = str(i)
    if i < 10:
        val = '00' + val
    if i >= 10 and i < 100:
        val = '0' + val
    save='plot_' + val + '.png'
    display.specular_plot(
        colorbar_flag=True, axis_label_flag=True, save=save,
        title='Minutes from 11/25/11 00 UTC: '+str([index_start, index_stop]),
        ms=25, cmap='gist_ncar', indices=[index_start, index_stop],
        lonrange=[50, 100], latrange=[-25, 25])

In [23]: merged_data = pygnss.CygnssMultiSat(output_L2_files, number=4)
display = pygnss.CygnssL2WindDisplay(merged_data)
multiple = 60
plot_animation(display, multiple=multiple)
```

0 60

No good specular points, not plotting

60 120

No good specular points, not plotting

120 180

No good specular points, not plotting

180 240

No good specular points, not plotting

240 300

No good specular points, not plotting

300 360

No good specular points, not plotting

360 420

No good specular points, not plotting

420 480

No good specular points, not plotting

480 540
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540 600
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600 660
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660 720
720 780
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780 840
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840 900
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900 960
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960 1020
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1020 1080
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1080 1140
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6900 6960
No good specular points, not plotting
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7020 7080
7080 7140
7140 7200
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7320 7380
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7440 7500
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7560 7620
7620 7680
7680 7740
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No good specular points, not plotting

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13620 13680
13680 13740
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13800 13860
13860 13920
13920 13980
13980 14040
14040 14100
14100 14160
14160 14220
14220 14280
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14340 14400
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14460 14520
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14580 14640
14640 14700
14700 14760
14760 14820
14820 14880
14880 14940
14940 15000
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16080 16140
16140 16200
16200 16260
16260 16320
16320 16380
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```

The basics of using pygnss to make a multi-panel plot ...

```

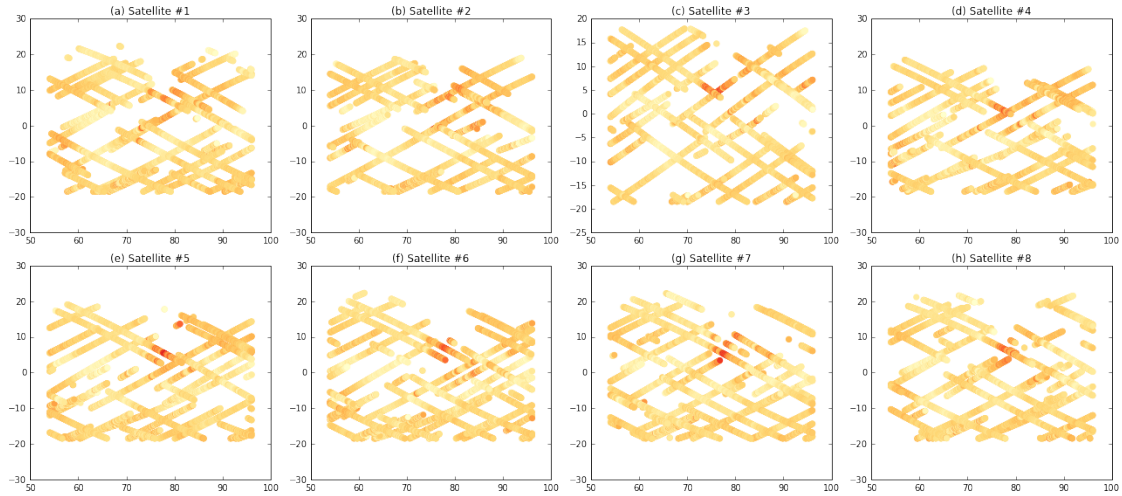
In [24]: fig = plt.figure(figsize=(18,8))
         dis = []
         for i, sat in enumerate(merged_data.satellites):
             dis.append(pygnss.CygnssL2WindDisplay(sat))
         ax = fig.add_subplot(241)
         dis[0].specular_plot(title='(a) Satellite #1')
         ax = fig.add_subplot(242)
         dis[1].specular_plot(title='(b) Satellite #2')
         ax = fig.add_subplot(243)
         dis[2].specular_plot(title='(c) Satellite #3')
         ax = fig.add_subplot(244)
         dis[3].specular_plot(title='(d) Satellite #4')
         ax = fig.add_subplot(245)

```

```

dis[4].specular_plot(title='(e) Satellite #5')
ax = fig.add_subplot(246)
dis[5].specular_plot(title='(f) Satellite #6')
ax = fig.add_subplot(247)
dis[6].specular_plot(title='(g) Satellite #7')
ax = fig.add_subplot(248)
dis[7].specular_plot(title='(h) Satellite #8')
plt.tight_layout()

```



1.0.6 Data Investigation

In this section I just show some uncommented examples of how to investigate the L2 wind data and input wind data within the pygnss framework.

L2 Wind Data Examples

```

In [25]: print(np.shape(merged_data.WindSpeed[1100:1300][:]))
          cond = merged_data.WindSpeed[1100:1300][:] >= 0
          print(merged_data.WindSpeed[1100:1300][:][cond])

```

```
(200, 120)
```

```
[]
```

```

In [26]: print(merged_data.satellites[3].RangeCorrectedGain[merged_data.satellites[3].GoodData == 1])

```

```
[ 0.          0.          0.          ..., 13.52930737 20.22728539
 23.73261642]
```

```

In [27]: cond = merged_data.WindSpeed >= 0
          print(merged_data.WindSpeed[cond])
          print(merged_data.TruthWindSpeed[cond])

```

```
[ 7.0156498  6.57197046  8.84980297 ..., 10.39377022  8.75372791
  9.71208382]
```

```
[ 7.90488243  7.90488243  7.89258528 ..., 7.05763531  7.07583094
  7.07583094]
```

```
In [28]: print(np.shape(merged_data.Longitude))
         indices=[180,240]
         print(np.shape(merged_data.Longitude[indices[0]:indices[1]][:]))
```

(86300, 120)

(60, 120)

Input Wind Data Examples

```
In [29]: print(input_data.latitude)
         print
         print(input_data.longitude)
```

```
[-18.6465683 -18.55637169 -18.46617508 -18.37598038 -18.28578377
-18.19558716 -18.10539055 -18.01519585 -17.92499924 -17.83480263
-17.74460602 -17.65441132 -17.56421471 -17.4740181 -17.38382149
-17.29362679 -17.20343018 -17.11323357 -17.02303696 -16.93284225
-16.84264565 -16.75244904 -16.66225243 -16.57205772 -16.48186111
-16.39166451 -16.3014679 -16.21127319 -16.12107658 -16.03087997
-15.94068336 -15.85048771 -15.76029205 -15.67009544 -15.57989883
-15.48970318 -15.39950752 -15.30931091 -15.2191143 -15.12891865
-15.03872299 -14.94852638 -14.85832977 -14.76813412 -14.67793751
-14.58774185 -14.49754524 -14.40734959 -14.31715298 -14.22695732
-14.13676071 -14.0465641 -13.95636845 -13.86617279 -13.77597618
-13.68577957 -13.59558392 -13.50538826 -13.41519165 -13.32499504
-13.23479939 -13.14460373 -13.05440712 -12.96421051 -12.87401485
-12.7838192 -12.69362259 -12.60342598 -12.51323032 -12.42303467
-12.33283806 -12.24264145 -12.15244579 -12.06225014 -11.97205353
-11.88185692 -11.79166126 -11.70146561 -11.611269 -11.52107239
-11.43087673 -11.34068012 -11.25048447 -11.16028786 -11.07009125
-10.97989559 -10.88969994 -10.79950333 -10.70930672 -10.61911106
-10.52891541 -10.4387188 -10.34852314 -10.25832653 -10.16813087
-10.07793427 -9.98773766 -9.897542 -9.80734539 -9.71714973
-9.62695312 -9.53675747 -9.44656086 -9.3563652 -9.26616859
-9.17597294 -9.08577633 -8.99558067 -8.90538406 -8.81518841
-8.7249918 -8.63479614 -8.54459953 -8.45440388 -8.36420727
-8.27401161 -8.183815 -8.09361935 -8.00342274 -7.91322708
-7.82303047 -7.73283482 -7.64263821 -7.55244255 -7.46224594
-7.37205029 -7.28185368 -7.19165802 -7.10146141 -7.0112648
-6.92106915 -6.83087254 -6.74067688 -6.65048027 -6.56028461
-6.47008801 -6.37989235 -6.28969574 -6.19950008 -6.10930347
-6.01910782 -5.92891121 -5.83871555 -5.74851894 -5.65832329
-5.56812668 -5.47793102 -5.38773441 -5.29753876 -5.20734215
-5.11714649 -5.02694988 -4.93675423 -4.84655762 -4.75636196
-4.66616535 -4.5759697 -4.48577309 -4.39557743 -4.30538082
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-2.8622427 -2.77204609 -2.68185043 -2.59165382 -2.50145721
-2.41126251 -2.3210659 -2.23086929 -2.14067268 -2.05047798
-1.96028137 -1.87008476 -1.77988815 -1.68969345 -1.59949684
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-1.05831909 -0.96812248 -0.87792778 -0.78773117 -0.69753456
-0.60733795 -0.51714325 -0.42694664 -0.33675003 -0.24655342
-0.15635872 -0.06616211 0.0240345 0.11423111 0.20442581]
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0.29462242	0.38481903	0.47501564	0.56521034	0.65540695	
0.74560356	0.83580017	0.92599487	1.01619148	1.10638809	
1.1965847	1.2867794	1.37697601	1.46717262	1.55736923	
1.64756584	1.73776054	1.82795715	1.91815376	2.00835037	
2.09854507	2.18874168	2.27893829	2.3691349	2.45932961	
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3.45148849	3.5416832	3.63187981	3.72207642	3.81227303	
3.90246773	3.99266434	4.08286095	4.17305756	4.26325226	
4.35344887	4.44364548	4.53384209	4.6240387	4.7142334	
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6.15737152	6.24756813	6.33776474	6.42796135	6.51815605	
6.60835266	6.69854927	6.78874588	6.87894058	6.96913719	
7.0593338	7.14953041	7.23972511	7.32992172	7.42011833	
7.51031494	7.60051155	7.69070625	7.78090286	7.87109947	
7.96129608	8.05149078	8.14168739	8.231884	8.32208061	
8.41227531	8.50247192	8.59266853	8.68286514	8.77305984	
8.86325645	8.95345306	9.04364967	9.13384438	9.22404099	
9.31423759	9.4044342	9.49462891	9.58482552	9.67502213	
9.76521873	9.85541344	9.94561005	10.03580666	10.12600327	
10.21619797	10.30639458	10.39659119	10.4867878	10.57698441	
10.66717911	10.75737572	10.84757233	10.93776894	11.02796364	
11.11816025	11.20835686	11.29855347	11.38874817	11.47894478	
11.56914139	11.659338	11.7495327	11.83972931	11.92992592	
12.02012253	12.11031723	12.20051384	12.29071045	12.38090706	
12.47110176	12.56129837	12.65149498	12.74169159	12.83188629	
12.9220829	13.01227951	13.10247612	13.19267273	13.28286743	
13.37306595	13.46326065	13.55345535	13.64365387	13.73384857	
13.82404327	13.91424179	14.00443649	14.09463501	14.18482971	
14.27502441	14.36522293	14.45541763	14.54561234	14.63581085	
14.72600555	14.81620407	14.90639877	14.99659348	15.08679199	
15.17698669	15.2671814	15.35737991	15.44757462	15.53777313	
15.62796783	15.71816254	15.80836105	15.89855576	15.98875427	
16.07894897	16.16914368	16.25934219	16.3495369	16.4397316	
16.52993011	16.62012482	16.71032333	16.80051804	16.89071274	
16.98091125	17.07110596	17.16130066	17.25149918	17.34169388	
17.4318924	17.5220871	17.6122818	17.70248032	17.79267502	
17.88286972	17.97306824	18.06326294	18.15346146	18.24365616	
18.33385086	18.42404938	18.51424408	18.60443878	18.6946373	18.784832
18.87503052	18.96522522	19.05541992	19.14561844	19.23581314	
19.32601166	19.41620636	19.50640106	19.59659958	19.68679428	
19.77698898	19.8671875	19.9573822	20.04758072	20.13777542	
20.22797012	20.31816864	20.40836334	20.49855804	20.58875656	
20.67895126	20.76914978	20.85934448	20.94953918	21.0397377	
21.1299324	21.22012711	21.31032562	21.40052032	21.49071884	
21.58091354	21.67110825	21.76130676	21.85150146	21.94169998	
22.03189468	22.12208939	22.2122879	22.3024826	22.39267731]	
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54.46520615	54.55527115	54.64533615	54.73540115	54.82546616	
54.91553116	55.00559616	55.09566116	55.18572617	55.27579117	
55.36585617	55.45592117	55.54598618	55.63604736	55.72611237	
55.81617737	55.90624237	55.99630737	56.08637238	56.17643738	

56.26650238	56.35656738	56.44663239	56.53669739	56.62676239	
56.71682739	56.8068924	56.8969574	56.9870224	57.0770874	
57.1671524	57.25721741	57.34728241	57.43734741	57.52741241	
57.61747742	57.70754242	57.79760742	57.88767242	57.97773743	
58.06780243	58.15786743	58.24793243	58.33799744	58.42806244	
58.51812744	58.60819244	58.69825745	58.78832245	58.87838745	
58.96845245	59.05851746	59.14858246	59.23864365	59.32870865	
59.41877365	59.50883865	59.59890366	59.68896866	59.77903366	
59.86909866	59.95916367	60.04922867	60.13929367	60.22935867	
60.31942368	60.40948868	60.49955368	60.58961868	60.67968369	
60.76974869	60.85981369	60.94987869	61.0399437	61.1300087	
61.2200737	61.3101387	61.4002037	61.49026871	61.58033371	
61.67039871	61.76046371	61.85052872	61.94059372	62.03065491	
62.12071991	62.21078491	62.30084991	62.39091492	62.48097992	
62.57104492	62.66110992	62.75117493	62.84123993	62.93130493	
63.02136993	63.11143494	63.20149994	63.29156494	63.38162994	
63.47169495	63.56175995	63.65182495	63.74188995	63.83195496	
63.92201996	64.01208496	64.10214996	64.19221497	64.28227997	
64.37234497	64.46240997	64.55247498	64.64253998	64.73260498	
64.82266998	64.91273499	65.00279999	65.09286499	65.18292999	
65.272995	65.36306	65.453125	65.54319	65.633255	
65.72332001	65.81338501	65.90345001	65.99351501	66.08358002	
66.17364502	66.26371002	66.35377502	66.44384003	66.53390503	
66.62397003	66.71403503	66.80409241	66.89416504	66.98422241	
67.07429504	67.16435242	67.25442505	67.34448242	67.43455505	
67.52461243	67.61467743	67.70474243	67.79480743	67.88487244	
67.97493744	68.06500244	68.15506744	68.24513245	68.33519745	
68.42526245	68.51532745	68.60539246	68.69545746	68.78552246	
68.87558746	68.96565247	69.05571747	69.14578247	69.23584747	
69.32591248	69.41597748	69.50604248	69.59610748	69.68617249	
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70.2265625	70.3166275	70.4066925	70.49675751	70.58682251	
70.67688751	70.76695251	70.85701752	70.94708252	71.03714752	
71.12721252	71.21727753	71.30734253	71.39740753	71.48747253	
71.57753754	71.66760254	71.75766754	71.84773254	71.93779755	
72.02786255	72.11792755	72.20799255	72.29805756	72.38812256	
72.47818756	72.56825256	72.65831757	72.74838257	72.83844757	
72.92851257	73.01856995	73.10864258	73.19869995	73.28877258	
73.37882996	73.46890259	73.55895996	73.64903259	73.73908997	
73.8291626	73.91921997	74.0092926	74.09934998	74.18942261	
74.27947998	74.36955261	74.45960999	74.54968262	74.63973999	
74.72980499	74.81987	74.909935	75.	75.090065	75.18013
75.27019501	75.36026001	75.45032501	75.54039001	75.63045502	
75.72052002	75.81058502	75.90065002	75.99071503	76.08078003	
76.17084503	76.26091003	76.35097504	76.44104004	76.53110504	
76.62117004	76.71123505	76.80130005	76.89136505	76.98143005	
77.07149506	77.16156006	77.25162506	77.34169006	77.43175507	
77.52182007	77.61188507	77.70195007	77.79201508	77.88208008	
77.97214508	78.06221008	78.15227509	78.24234009	78.33240509	
78.42247009	78.5125351	78.6026001	78.6926651	78.7827301	
78.8727951	78.96286011	79.05292511	79.14299011	79.23305511	
79.32312012	79.41318512	79.50325012	79.5933075	79.68338013	
79.7734375	79.86351013	79.9535675	80.04364014	80.13369751	
80.22377014	80.31382751	80.40390015	80.49395752	80.58403015	

80.67408752	80.76416016	80.85421753	80.94429016	81.03434753
81.12442017	81.21447754	81.30454254	81.39460754	81.48467255
81.57473755	81.66480255	81.75486755	81.84493256	81.93499756
82.02506256	82.11512756	82.20519257	82.29525757	82.38532257
82.47538757	82.56545258	82.65551758	82.74558258	82.83564758
82.92571259	83.01577759	83.10584259	83.19590759	83.2859726
83.3760376	83.4661026	83.5561676	83.6462326	83.73629761
83.82636261	83.91642761	84.00649261	84.09655762	84.18662262
84.27668762	84.36675262	84.45681763	84.54688263	84.63694763
84.72701263	84.81707764	84.90714264	84.99720764	85.08727264
85.17733765	85.26740265	85.35746765	85.44753265	85.53759766
85.62766266	85.71772766	85.80779266	85.89785767	85.98792267
86.07798767	86.16804504	86.25811768	86.34817505	86.43824768
86.52830505	86.61837769	86.70843506	86.79850769	86.88856506
86.9786377	87.06869507	87.1587677	87.24882507	87.33889771
87.42895508	87.51902771	87.60908508	87.69915771	87.78921509
87.87928772	87.96934509	88.05941772	88.1494751	88.23954773
88.3296051	88.41967773	88.50973511	88.59980774	88.68986511
88.77993011	88.86999512	88.96006012	89.05012512	89.14019012
89.23025513	89.32032013	89.41038513	89.50045013	89.59051514
89.68058014	89.77064514	89.86071014	89.95077515	90.04084015
90.13090515	90.22097015	90.31103516	90.40110016	90.49116516
90.58123016	90.67129517	90.76136017	90.85142517	90.94149017
91.03155518	91.12162018	91.21168518	91.30175018	91.39181519
91.48188019	91.57194519	91.66201019	91.7520752	91.8421402
91.9322052	92.02226257	92.11233521	92.20239258	92.29246521
92.38252258	92.47259521	92.56265259	92.65272522	92.74278259
92.83285522	92.9229126	93.01298523	93.1030426	93.19311523
93.28317261	93.37324524	93.46330261	93.55337524	93.64343262
93.73350525	93.82356262	93.91363525	94.00369263	94.09376526
94.18382263	94.27389526	94.36395264	94.45402527	94.54408264
94.63415527	94.72421265	94.81428528	94.90434265	94.99441528
95.08447266	95.17454529	95.26460266	95.35466766	95.44473267
95.53479767	95.62486267	95.71492767	95.80499268	95.89505768
95.98512268				

```
In [30]: print(np.max(input_data.latitude), np.min(input_data.latitude))
          print(np.max(input_data.longitude), np.min(input_data.longitude))
```

```
22.3927 -18.6466
95.9851 54.0149
```

```
In [31]: print(input_data.time, np.shape(input_data.northward_wind))
```

```
[ -6.18198876e+10] (1, 456, 467)
```

```
In [32]: print(np.shape(input_data.WindSpeed), np.shape(input_data.longitude),
              np.shape(input_data.latitude))
```

```
(1, 456, 467) (467,) (456,)
```

```
In [ ]:
```